Sora Kim

HCI Researcher · Human-AI Interaction and Accountability in High-Stakes Systems

sorakim.lab@gmail.com (+82) 10-8119-7439 sorakim-lab.github.io

RESEARCH FOCUS

I study how accountability is structured in regulated work and how system design shapes which causal explanations remain available before responsibility settles — a research program I develop under the name *Regulatory UX*. My research combines HCI experimentation with STS analysis, using pharmaceutical GMP environments as a primary empirical site. Intervention work examines human-AI interaction as a reasoning intervention: resisting premature closure and preserving the comparability of alternative explanations before attribution stabilizes.

RESEARCH

HCI

Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations

Preprint, 2026 · CHI 2027 full paper submission in preparation · CSCW 2026 poster under review

Two-study experimental design (N=20 + N=20). Study 1 shows time-sensitive convergence: 15/20 participants produced single-cause explanations initially, while 18/20 participants broadened their accounts within 10 minutes. Study 2 introduces an AI-supported reasoning scaffold enforcing explicit comparison, with 15/20 participants shifting attribution and demonstrating increased causal breadth.

From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work

Working paper, 2026 · DIS 2027 submission in preparation

Think-aloud study with N=20 over a 500-document synthetic SOP corpus. Reduces retrieval time from $\sim$6 minutes to $\sim$1 minute and shows that access conditions expand the range of procedures considered actionable, shaping interpretation upstream of task execution.

When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation

Working paper, 2026 · CSCW 2027 submission in preparation

Two-study prototype evaluation (N=20). Introduces loopback behavior as an empirical indicator of document-structural friction and demonstrates that navigation difficulty reflects missing or non-navigable document relationships rather than user error.

Friction Before Failure: Making Review Burden Visible in Regulated Document Work

Working paper, 2026 · CSCW 2027 full paper submission in preparation

Prototype-based study (N=20) introducing an external observability layer for deviation–CAPA workflows. Shows that making coordination work visible enables participants to interpret delay as structural rather than individual, without automating quality decisions.

How Regulated Systems Produce Responsibility: A Four-Layer Framework for Interaction Design

Working paper, 2026 · TOCHI submission in preparation

Synthesizes four empirical HCI studies into a unified framework (Regulatory UX) explaining how system design structures accountability conditions across access, structure, coordination, and reasoning layers.

STS — Theoretical Foundations

Anticipated Accountability Convergence: Structuring Responsibility Before Action
Preprint, 2026
Argues that accountability expectations are prospectively structured through procedural rules, documentation regimes, and evaluative infrastructures — shaping conduct before any deviation or investigation occurs.

Structuring Judgment Under Constraint: Cognitive Friction in Procedural Texts
Preprint, 2026
Examines how procedural text structure — conditional branching, cross-references, discretionary qualifiers — creates localized judgment demands during execution, distributing interpretive burden unevenly across a workforce stratified by experience.

Responsibility Convergence in Organizational Investigations
Preprint, 2026
Analyzes how investigations under closure pressure settle around fast, documentable explanations and identifiable actors — establishing the retrospective stabilization mechanism that the HCI scaffolding work is designed to resist.

Premature Accountability Convergence in Regulated Organizations
Preprint, 2026 · 4S Toronto 2026 *under review*
Introduces PAC as a structural acceleration dynamic: AI-mediated representations intensify pre-existing convergence tendencies by compressing interpretive space earlier in the sensemaking sequence, before alternative causal accounts remain viable. Positioned within STS discussions on accountability, documentation, and organizational analysis.

From GMP to High-Stakes Systems: Accountability Convergence as a Cross-Domain Phenomenon in Regulated Investigations
Manuscript in preparation, 2026
Extends the AAC–CF–RC–PAC framework beyond pharmaceutical manufacturing, examining how the same convergence mechanisms operate across healthcare and aviation investigation documents — grounding generalizability claims in publicly available regulatory and incident reports.

PROFESSIONAL EXPERIENCE

Quality Control / Quality Systems Project Lead Sep 2022 – Present
Prestige Biologics, South Korea

- Led quality system operations across a 30-person QC/QA team, managing 20+ deviation and CAPA investigations end-to-end in GMP-regulated pharmaceutical manufacturing.
- Supported 10+ regulatory inspections and global audits; developed working expertise in how documentation architecture shapes traceability, coordination, and responsibility attribution.
- Revised and maintained 50+ SOPs across validation, deviation, CAPA, and RCA workflows, directly observing how procedural document design produces accountability patterns, informing my research on accountability in regulated investigations.
- Designed and implemented the organization's first documentation framework for pharmaceutical technology transfer — including master plans, reports, and gap analysis protocols — establishing traceability and accountability structures where none previously existed.

METHODS & SKILLS

Methods	Experimental design (within-subject, task-based); think-aloud protocol; behavioral logging; causal reasoning analysis; annotation protocol design; qualitative coding (open, axial); document and corpus analysis; NLP / text processing (TF-IDF, sentence-level annotation)
Tools	Python · R · Streamlit · LaTeX · HTML/CSS · GitHub
Languages	Korean (native) · English (fluent) · Japanese (fluent)